FHIR API Data Ingestion & Analytics Assignment

🔗 Prerequisites

1. Environment Setup

- Access to **Microsoft Fabric** or **Databricks** is required.
- If you don't have Fabric:
  - Create a **free 60-day Microsoft Fabric trial** using your personal Microsoft account.
  - Alternatively, use **Databricks (free tier)** for development.

2. Source Control

- Use **GitHub** or **Azure DevOps Repos** for version control.
- Submit your work by:
  - Sharing the Git repository link, or
  - Uploading a ZIP file of the project.

Task 1: Incremental Ingestion of FHIR API Data

📎 API Documentation

Use the public FHIR API:
🔲 https://hapi.fhir.org/baseR4/swagger-ui/

Objective: Build a Medallion Lakehouse Architecture

Analytical Schema (Core Tables)

Ingest and model the following FHIR resources:

- Patient
- Encounter
- Observation
- Condition

Data Ingestion Requirements

Use **Spark Notebooks** to:

- Fetch data incrementally for **2–3 days** using **pagination**.
- Store **JSON / XML Response** as it is in **Lakehouse**
- Convert **JSON / XML responses** into **Delta Tables** or **Parquet** format.
- Store data in the **Bronze Layer** of the Lakehouse.

Add Metadata Columns:

- extraction_timestamp
- api_url_or_params

Implement Data Versioning:

- Track:
    - When each API was called.
    - When data was saved.
    - Check if any data was changed daily loads and maintain historical versions using Slowly Changing Dimension Type 2 (SCD Type 2) logic.

Data Transformation & Modeling

Medallion Architecture Layers

1. **Raw Layer**
    - Store API responses bucketed by date.
    - Maintain folder structure for all API objects.
2. **Bronze Layer**
    - Store raw ingested data in Delta/Parquet format.
3. **Silver Layer**
    - Clean and deduplicate data using **Spark** or **Dataflows Gen2**.
4. **Gold Layer**
    - Create final warehouse views optimized for **reporting and analytics**.

⚠ Use **Dataflow Gen2** only for transformations, not for JSON-to-table conversion.

Data Orchestration

Build a pipeline using **Fabric Data Pipeline** or **Databricks Workflow** to:

- Orchestrate API calls in the following order:
  Patient → Encounter → Observation → Condition
- Handle:
  - Transformation
  - Metadata logging
  - Data loading

✅ Submission Checklist

- Ingest and transform **2–3 days of data** with pagination.
- Create **Medallion Architecture** (Raw → Bronze → Silver → Gold).
- Add **metadata** and **versioning** to all tables.
- Ensure **modular and reusable code** (no hardcoding).
- Submit via **Git repo** or **ZIP file** with:
  - All code
  - Notebooks
  - Pipeline setup
- Include **clear documentation** of pipeline and table relationships.

Optional Extensions

- Repeat the task using **XML** format instead of JSON.
- Build a **1-page Power BI report** using the **Gold Layer**.